

A new density limit for unanimity in majority dynamics on random graphs

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Abstract

Majority dynamics is a process on a simple, undirected graph G with an initial Red/Blue color for every vertex of G . Each day, each vertex updates its color following the majority among its neighbors, using its previous color for tie-breaking. The dynamics achieves *unanimity* if every vertex has the same color after finitely many days, and such color is said to *win*.

When G is a $G(n, p)$ random graph, L. Tran and Vu (2019) found a condition in terms of p and the initial difference 2Δ between the sizes of the Red and Blue camps, such that unanimity is achieved with probability arbitrarily close to 1. They showed that if $p\Delta^2 \gg 1$, $p\Delta \geq 100$, and $p \geq (1 + \varepsilon)n^{-1} \log n$ for a positive constant ε , then unanimity occurs with probability $1 - o(1)$. If p is not extremely small, namely $p > \log^{-1/16} n$, then Sah and Sawhney (2022) showed that the condition $p\Delta^2 \gg 1$ is sufficient.

If $n^{-1} \log^2 n \ll p \ll n^{-1/2} \log^{1/4} n$, we show that $p^{3/2}\Delta \gg n^{-1/2} \log n$ is enough. Since this condition holds if $p\Delta \geq 100$ for p in this range, this is an improvement of Tran's and Vu's result. For the closely related problem of finding the optimal condition for p to achieve unanimity when the initial coloring is chosen uniformly at random among all possible Red/Blue assignments, our result implies a new lower bound $p \gg n^{-2/3} \log^{2/3} n$, which improves upon the previous bound of $n^{-3/5} \log n$ by Chakraborti, Kim, Lee and T. Tran (2021).

1 Introduction

1.1 Majority Dynamics

Consider a parliamentary election with two parties. The election season begins with Day 0, where each voter supports one side. At the end of every day, each voter polls their friends, then supports the majority side the next morning, or keeps their affiliation in case of a tie. On election day, everyone votes for the party they currently supports and the parliament will be divided proportionally with each side's number of votes.

This process can be modeled with a graph process, where vertices represent voters and edges represent friendships. We let Red and Blue, or R and B, stand for the two parties. The state of affiliations at any time is simply a coloring on the graph. This process on a graph is called **Majority Dynamics** in literature, due to the majority-based updating rule.